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Knife insert and retract with independent knife protection of agricultural baler

Abstract

An agricultural baler includes: a frame; a feeder system coupled with the frame and including: a cutting assembly coupled with the frame and including: at least one knife configured for cutting a crop material; an engagement apparatus configured for: being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife; selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a first position; and selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a second position.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention pertains to an agricultural baler, and, more specifically, to a crop cutting assembly of the agricultural baler.

BACKGROUND OF THE INVENTION

(2) Agricultural harvesting machines, such as agricultural balers (which can be referred to as balers), have been used to consolidate and package crop material (which, depending upon the application, can also be referred to as forage, forage material, or forage crop material) so as to facilitate the storage and handling of the crop material for later use. Often, a mower-conditioner

cuts and conditions the crop material for swath or windrow drying in the sun. When the cut crop material is properly dried (depending upon the application), an agricultural harvesting machine, such as an agricultural baler, which can be a round baler, travels along the swath or windrows (hereinafter, collectively referred to as windrows, unless otherwise specified) to pick up the crop material. In the case of round balers, the crop material is formed into cylindrically-shaped round bales.

(3) More specifically, pickups of the baler gather the cut and windrowed crop material from the ground, and then convey the cut crop material toward a bale-forming chamber within the baler (that is, the bale chamber). A drive mechanism operates to activate any pickups, augers, and/or a rotor of a feed mechanism (which can also be referred to as a feeder system). A pickup can convey crop material in an overshot manner, while a rotor can convey crop material toward or into the bale chamber in an undershot manner. A conventional bale chamber may include a pair of opposing sidewalls with a series of rolls (which can be referred to as rollers) and belts that rotate and compress the crop material into a cylindrical shape. When the bale has reached a desired size and density, a wrapping assembly, which includes wrap material, may wrap the bale to ensure, at least in part, that the bale maintains its shape and density. The wrap material can include a film (such as a flexible plastic wrap) or a net (which can be referred to as net wrap). For example, wrap material may be used to wrap the bale of crop material. After wrapping, a cutting or severing mechanism of the wrapping assembly may be used to cut the wrap material once the bale has been wrapped. The wrapped bale may be ejected from the baler and onto the ground by, for example, raising a tailgate of the baler. The tailgate is then closed, and the cycle repeated as necessary and desired to manage the field of cut crop material.

(4) The feeder system can include not only the rotor but also a floor and a cutting assembly. The rotor, which is downstream of the pickup, can be positioned above the floor which the crop material traverses prior to entering the bale chamber and can work in conjunction with, and cooperatively with, the cutting assembly. The rotor can include a rotor shaft (extending transversely) and a plurality of tines (which can have a generally triangular or star-shaped configuration) spaced apart across the transverse extent of the baler. The tines are configured to engage and thereby to push the crop material towards the bale chamber and can be grouped in pairs, with a relatively short distance between the tines of a given pair. The floor can include a plurality of slots across the transverse extent of the floor, each slot extending longitudinally in the floor (that is, parallel or otherwise aligned with a direction of crop flow). The cutting assembly can include a plurality of knives (which can also be referred to as cutters) which are selectively received in the slots of this floor, respectively. When inserted through the slots so that the knives extend at least partially above the floor, each respective knife (depending upon the design) can extend between a pair of tines of the rotor, as the rotor shaft rotates the tines. Further, the knives, as they extend through the slots above the floor are configured to cut the crop material to a predetermined length, as the crop material passes by the knives prior to the crop material entering the bale chamber. By cutting the crop material into smaller lengths prior to entering the bale chamber, a denser bale can be formed in the bale chamber, which advantageously provides more crop material per bale, enables less wrap material to be used to wrap the bale, and enables better stacking of bales during storage and/or transit.

(5) When a foreign object, such as a rock, is taken up by the pickup and conveyed towards the bale chamber, the foreign object can encounter one or more knives. To prevent or otherwise mitigate damage to the knives, the knives can have overload protection by way of an overload protection mechanism, which can include a spring which urges one or more knives to extend at least partially above the floor and allows the one or more knives to drop down at least partially below the floor when the foreign object strikes the one or more knives. Individual knife overload protection is known to use a spring for each knife. Having such overload protection for each knife (which can be referred to as individual knife overload protection) is advantageous. Overload protection is also

known, not using springs, that is not assigned to individual knives but to an entire bank of knives of the feeder system.

(6) During use, knives can become jammed or otherwise stuck with respect to the slots in the floor, which can inhibit servicing and/or replacing of respective knives. That is, dirt and debris can collect around or enter into the slots, thereby causing at least two problems. First, knives already deployed at least partially above the floor can become stuck in their respective slots because of the dirt and debris and thus unable to retract back down through the slots upon encountering a foreign object or otherwise needing to be retracted or serviced. Second, knives not yet deployed at least partially above the floor can become unable to progress through the respective slots because of the dirt and debris that has collected at the slots and/or on the knives, blocking the knives from deploying through the slots into position for cutting; for, the spring force urging the knives into a deployed position (and thus providing the overload protection) is not strong enough to burst through this blockage. To clear a blockage, for example, that is preventing an individual knife from inserting up through the slot, it is known to use a hydraulic linear actuator to push on the spring of the individual knife overload protection mechanism; but, this design is often not strong enough to push through the blockage.

(7) In the current state of the art, feeder systems include either an individual knife overload protection mechanism with a hydraulic linear actuator to push through a spring of the individual knife overload protection mechanism to clear a blockage, or the ability to force all knives to be inserted or retracted. Neither option is fully satisfactory, and the former can be improved upon.

(8) What is needed in the art is a simple and effective way to have, simultaneously, both individual knife overload protection and an effective way to forcibly insert and to forcibly retract the knives of a feeder system.

SUMMARY OF THE INVENTION

(9) The present invention provides an agricultural baler with a feeder system including individual knife overload protection that is spaced apart from an engagement apparatus configured for forcibly inserting and retracting all or a group of knives of the feeder system.

(10) The invention in one form is directed to a feeder system of an agricultural baler, the feeder system being coupled with a frame of the agricultural baler, the feeder system including: a cutting assembly coupled with the frame and including: at least one knife configured for cutting a crop material; an engagement apparatus configured for: being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife; selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a first position; and selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a second position.

(11) The invention in another form is directed to an agricultural baler, including: a frame; a feeder system coupled with the frame and including: a cutting assembly coupled with the frame and including: at least one knife configured for cutting a crop material; an engagement apparatus configured for: being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife; selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a first position; and selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a second position.

(12) The invention in yet another form is directed to a method for using an agricultural baler, the method including the steps of: providing a frame and a feeder system coupled with the frame, the feeder system including a cutting assembly coupled with the frame and including at least one knife and an engagement apparatus, the at least one knife configured for cutting a crop material, the engagement apparatus being configured for being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife; engaging selectively, by way of the engagement apparatus, with the at least one knife and thereby forcing the

at least one knife to occupy a first position; and engaging selectively, by way of the engagement apparatus, with the at least one knife and thereby forcing the at least one knife to occupy a second position.

(13) An advantage of the present invention is that it provides individual knife overload protection.

(14) Another advantage is that it provides a way to forcibly insert and retract the knives. This may be done to clear blockage of the knives due to dirt and debris, to raise or lower the knives for servicing, or to run the baler with the knives down so as not to cut the crop material.

(15) Yet another advantage is that it provides a way to have individual knife overload protection employing a spring while being able to selectively engage and thereby force the knives up or down. Thus, a mechanism is provided that forces knives to insert and to retract when needed (that is, the knives are positively inserted and positively retracted) and that allows a respective knife to fall away in an overload event.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For the purpose of illustration, there are shown in the drawings certain embodiments of the present invention. It should be understood, however, that the invention is not limited to the precise arrangements, dimensions, and instruments shown. Like numerals indicate like elements throughout the drawings. In the drawings:

(2) FIG. 1 illustrates a schematic side view of an exemplary embodiment of an agricultural vehicle, formed as a tractor, and an agricultural baler, the agricultural baler including a feeder system, in accordance with an exemplary embodiment of the present invention;

(3) FIG. 2 illustrates a perspective view of the feeder system of the agricultural baler of FIG. 1, the feeder system including a cutting assembly including a knife, with portions broken away, in accordance with an exemplary embodiment of the present invention;

(4) FIG. 3 illustrates a schematic perspective view of the cutting assembly of FIG. 2, with portions broken away, the cutting assembly including an engagement apparatus, in accordance with an exemplary embodiment of the present invention;

(5) FIG. 4 illustrates a schematic side view of the cutting assembly with the engagement apparatus of FIG. 3, with portions broken away, the engagement apparatus being in a home position and the knife being in a normal operating position, in accordance with an exemplary embodiment of the present invention;

(6) FIG. 5 illustrates a schematic side view of the cutting assembly with the engagement apparatus of FIG. 3, with portions broken away, the engagement apparatus being in a first position, which forcibly places the knife in an insertion position, in accordance with an exemplary embodiment of the present invention;

(7) FIG. 6 illustrates a schematic side view of the cutting assembly with the engagement apparatus of FIG. 3, with portions broken away, the engagement apparatus being in a second position, which forcibly places the knife in a retraction position, in accordance with an exemplary embodiment of the present invention;

(8) FIG. 7 illustrates a schematic side view of the cutting assembly with a second embodiment of the engagement apparatus, with portions broken away, in accordance with an exemplary embodiment of the present invention; and

(9) FIG. 8 illustrates a flow diagram showing a method for using an agricultural baler, in accordance with an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(10) The terms “forward”, “rearward”, “left” and “right”, when used in connection with the agricultural vehicle, agricultural baler, and/or components thereof are usually determined with

reference to the direction of forward operative travel of the agricultural vehicle and/or agricultural baler, but they should not be construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the agricultural vehicle and/or agricultural baler and are equally not to be construed as limiting. The terms “downstream” and “upstream” are determined with reference to the intended direction of crop material flow during operation, with “downstream” being analogous to “rearward” and “upstream” being analogous to “forward.”

(11) Referring now to the drawings, and more particularly to FIG. 1, there is shown an embodiment of an agricultural vehicle **100** (which can be referred to as a work vehicle **100**) towing an agricultural baler **101**, in accordance with the present invention, to perform a baling operation within a field. As shown, work vehicle **100** can be configured as an agricultural tractor, such as an operator-driven tractor or an autonomous tractor. However, in some embodiments, the work vehicle **100** may correspond to any other suitable vehicle configured to tow a baler across a field or that is otherwise configured to facilitate the performance of a baling operation, including an autonomous baling vehicle. Additionally, as shown, baler **101** can be configured as a round baler configured to generate round bales. However, in some embodiments, baler **101** may have any other suitable configuration, including being configured to generate square or rectangular bales. It should be further appreciated that baler **101**, while shown as being towed by tractor **100**, may also be a self-propelled baler that does not rely on a separate vehicle for propulsion and/or power to function.

(12) Work vehicle **100** includes a pair of front wheels **102**, a pair of rear wheels **103**, and a chassis **104** coupled to and supported by the wheels **102**, **103**. An operator's cab **105** may be supported by a portion of the chassis **104** and may house various input devices for permitting an operator to control the operation of work vehicle **100** and/or baler **101**. Additionally, work vehicle **100** may include an engine and a transmission mounted on chassis **104**. The transmission may be operably coupled to the engine and may provide variably adjusted gear ratios for transferring engine power to wheels **103** via a drive axle assembly.

(13) As shown in FIG. 1, work vehicle **100** may be coupled to baler **101** via a power take-off (PTO) **106** and a tongue **107** to a hitch of work vehicle **100** to allow vehicle **100** to tow baler **101** across the field. As such, work vehicle **100** may, for example, guide baler **101** toward crop material **136** deposited in windrows on the field. As is generally understood, to collect the crop material **136**, baler **101** includes a feeder system (which can be referred to generally as a crop collector) mounted on a front end of baler **101**. Feeder system **108** may, for example, include a pickup **111**, a rotor **121**, and a cutting assembly **128**. Pickup **111** includes a rotating wheel with tines that collect crop material **136** from the ground and direct the crop material **136** toward a bale chamber **109** of baler **101** in an overshot manner (rotating clockwise in FIG. 1). Rotor **121** includes a rotating shaft (rotor shaft) and a plurality of generally triangular or star-shaped tines (rotor tines) mounted to the shaft (as described above) that push or otherwise move crop material **136** towards or into bale chamber **109**, in an undershot manner (rotating counter-clockwise in FIG. 1). Feeder system **108** can also include a rotating shaft (not shown) generally between pickup and rotor **121** that includes side augers for moving crop material **136** inwardly prior to entering bale chamber **109**. Cutting assembly **128** is disposed generally below rotor **121** and includes a floor **131** and a plurality of knives **129**.

(14) Inside bale chamber **109**, rollers, belts, and/or other devices compact the crop material **136** to form a generally cylindrically-shaped bale **110**. Bale **110** is contained within baler **101** until ejection of bale **110** is instructed (e.g., by the operator and/or a baler controller **123** of baler **101**). In some embodiments, bale **110** may be automatically ejected from baler **101** once bale **110** is formed, by baler controller **123** detecting that bale **110** is fully formed and outputting an appropriate ejection signal. Further, work vehicle **100** includes a control system **114**, which includes a controller **115**, which includes a processor **116**, memory **117**, data **118**, and instructions **119**. Control system **114** can further include an input/output device **120** such as a laptop computer (with

keyboard and display) or a touchpad (including keypad functionality and a display), device **120** being configured for a user to interface therewith.

(15) As shown in FIG. **1**, baler **101** may also include a tailgate **112** movable between a closed position (as shown in the illustrated embodiment) and an opened position via a suitable actuator assembly. Tailgate **112** and/or the actuator assembly may be controlled to open and close by baler controller **123**. In the closed position, tailgate **112** may confine or retain bale **110** within baler **101**. In the open position, tailgate **112** may rotate out of the way to allow bale **110** to be ejected from the bale chamber **109**. Additionally, as shown in FIG. **1**, baler **101** may include a ramp **113** extending from its aft end that is configured to receive and direct bale **110** away from baler **113** as it is being ejected from bale chamber **109**. In some embodiments, ramp **113** may be spring loaded, such that ramp **113** is urged into a raised position, as illustrated. In such embodiments, the weight of bale **110** on ramp **113** may drive ramp **113** to a lowered position in which ramp **113** directs bale **110** to the soil surface. Once bale **110** is ejected, bale **110** may roll down ramp **113** and be deposited onto the field. As such, ramp **113** may enable bale **110** to maintain its shape and desired density by gently guiding bale **110** onto the field. Further, baler **101** includes a control system **122**, which includes controller **123**, which includes a processor **124**, memory **125**, data **126**, and instructions **127**. Controller **123** can communicate with controller **115**, so that controller **115** outputs information to the display of input/output device **120** of work vehicle **100**, thereby informing a user of various conditions of baler **101** and bales **110** forming or formed therein. Further, baler **101** includes a frame **130** to which all of the components of baler **101** are directly or indirectly coupled. Thus, feeder system **108**, and thus also cutting assembly **128**, are coupled with frame **130**.

(16) It should be appreciated that the configuration of work vehicle **100** described above and shown in FIG. **1** is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of work vehicle configuration. For example, in an alternative embodiment, a separate frame or chassis may be provided to which the engine, transmission, and drive axle assembly are coupled, a configuration common in smaller tractors. Still other configurations may use an articulated chassis to steer work vehicle, or rely on tracks in lieu of wheels **102**, **103**. Additionally, as indicated previously, work vehicle **100** may, in some embodiments, be configured as an autonomous vehicle. In such embodiments, work vehicle **100** may include suitable components for providing autonomous vehicle operation and, depending on the vehicle configuration, need not include the operator's cab **105**.

(17) Additionally, it should be appreciated that the configuration of baler **101** described above and shown in FIG. **1** is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of baler configuration. For example, as indicated previously, baler **101** may, in some embodiments, correspond to a square baler configured to generate square or rectangular bales. It should be further appreciated that the illustration of baler **101** in FIG. **1** is schematic.

(18) Further, in general, controllers **115**, **123** may each correspond to any suitable processor-based device(s), such as a computing device or any combination of computing devices. Each controller **115**, **123** may generally include one or more processor(s) **116**, **124** and associated memory **117**, **125** configured to perform a variety of computer-implemented functions (e.g., performing the methods, steps, algorithms, calculations and the like disclosed herein). Thus, each controller **115**, **123** may include a respective processor **116**, **124** therein, as well as associated memory **117**, **125**, data **118**, **126**, and instructions **119**, **127**, each forming at least part of the respective controller **115**, **123**. As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the respective memory **117**, **125** may generally include memory element(s) including, but not limited to, computer readable medium (e.g., random access memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a

compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD), and/or other suitable memory elements. Such memory **117, 125** may generally be configured to store information accessible to the processor(s) **116, 124**, including data **118, 126** that can be retrieved, manipulated, created, and/or stored by the processor(s) **116, 124** and the instructions **119, 127** that can be executed by the processor(s) **116, 124**. In some embodiments, data **118, 126** may be stored in one or more databases.

(19) Baler controller **123**, herein, is assumed to be the primary controller for controlling operations of baler **101**. However, controller **123**, as indicated in FIG. 1, can be in communication with controller **115** of work vehicle **100**, such that any or all information associated with either controller **115, 123** can be shared with the other controller **115, 123**, and either controller **115, 123** can perform the functions of the other controller **115, 123**. Controllers **115, 123** can communicate with each other in any suitable manner, such as a wired connection or a wireless connection, such as radio signals (RF), light signals, cellular, WiFi, Bluetooth, Internet, via cloud-based devices such as servers, and/or the like. Further, while not shown, both controllers **115, 123** can communicate with a remotely located data center, which controllers **115, 123** can communicate with by any suitable way, such as those just referenced. Such a data center can include its own controller (and thus processor(s), memory, data, and instructions, substantially similar to that described above with respect to controllers **115, 123**) which can be configured to perform any of the functions associated with controllers **115, 123**. Controllers **115, 123** and the data center can be a part of any network facilitating such communication therebetween, such as a local area network, a metropolitan area network, a wide area network, a neural network, whether wired or wireless.

(20) Referring now to FIG. 2, there is shown a perspective view of feeder system **108**, with portions broken away. Shown are pickup **111** (with tines), rotor **121**, and cutting assembly **128**. Rotor **121** is shown to include rotor shaft **233** and rotor tines **234** mounted to rotor shaft **233**. Rotor tines **234** are spaced apart transversely from one another and, as shown in FIG. 2, can be grouped in pairs. As rotor shaft **233** rotates, a given pair of rotor tines receive therebetween a respective upstanding knife **129**. Cutting assembly **128** is shown to include floor **131** and a plurality of knives **129** extending transversely across the front of baler **101**. Floor **131** includes a plurality of longitudinally extending slots **232** (running generally in the flow direction of crop material **136**) through which a respective knife **129** can extend when deployed. That is, each slot **232** allows a respective knife **129** to project and to pass therethrough; any suitable number of slots **232** can be provided in floor **131**. FIG. 2 shows several such knives **129** already having been inserted through respective slots and thus deployed and ready to encounter crop material **136**. Knives **129** are configured for cutting crop material **136** as crop material **136** is urged in the direction of flow **237** of crop material **136**. Knives **129** cut crop material **136** to a predetermined length, such as two-and-one-half inches, for example and not by way of limitation.

(21) Referring now to FIG. 3, there is shown a perspective view of a portion of feeder system **108**, which includes cutting assembly **128**. Cutting assembly **128** includes floor **131**, knives **129** (not shown in FIG. 3), and further includes an engagement apparatus **340**. Engagement apparatus **340** includes a pivot bar **341**, an engagement rod **342**, a joining mechanism **343**, and an actuator **344**. Floor **131** includes slots **232** and is coupled with frame **130** (as shown schematically in FIG. 3). Pivot bar **341** (which can be formed as a tube) can be rotatably coupled with frame **130** so as to pivot about pivot axis **345** in either direction, as indicated by double-arrow **347**. Pivot bar **341** can be made of any suitable material, such as steel. Pivot bar **341**, as shown in FIG. 3, can extend the transverse width of floor **131** and beyond (or, alternatively, within) the lateral sides of floor **131**. Engagement rod **342** (which can be formed as a tube) can be made of any suitable material, such as steel. Engagement rod **342** extends substantially parallel to pivot bar **341** and can be substantially the same length as pivot bar **341**. Joining mechanism **343** can be formed as a linkage **343**, as shown in FIG. 3, and can be made of any suitable material, such as steel. Two such linkages **343** are provided and disposed at each lateral end of pivot bar **341** and engagement rod **342**, such that each

linkage **343** is fixedly attached (such as by welding, pinning, or the like) to respective lateral ends of pivot bar **341** and engagement rod **342**. Linkages **343** pivot together with pivot bar **341**, causing engagement rod **342** to revolve around pivot bar **341** (thus, substantially in a circle) and thus be angularly displaced about pivot bar **341**; this angular displacement of engagement rod **342** relative to pivot bar **341** is only partial—less than 180 degrees (for example, 170 degrees). The limitation of such angular displacement can be controlled by actuator **344** and/or any associated control system. Actuator **344**, as shown in FIG. **3**, can be formed as a linear actuator **344**, such as a cylinder assembly **344**, including a cylinder **348** (housing a piston) and a rod **349** that is moved linearly by the action of the piston. As a linear actuator, cylinder assembly **344** can be, for example, a hydraulic cylinder assembly, a pneumatic cylinder assembly, or an electrically actuated cylinder assembly (which can include gearing and/or screws). If cylinder assembly **344** is a hydraulic cylinder assembly **344**, for example, then cylinder assembly can be controlled by suitable valving and can include fluid lines which fluidly communicate an interior of cylinder **348** with a pump and a reservoir of hydraulic cylinder fluid, such as hydraulic oil; this pump and reservoir can be maintained on work vehicle **100** and/or on baler **101**.

(22) FIG. **3** shows that two such actuators **344** are provided, each associated with respective lateral ends of pivot bar **341** and engagement rod **342**, as well as a respective linkage **343**. A proximal end of cylinder **348** (left-most end in FIG. **3**) is pivotably mounted, so as to be coupled with frame **130** (this coupling with frame **130** is shown schematically in FIG. **3**); cylinder **348** pivots about a pivot axis **350**. Cylinder **340** pivots about pivot axis **350** as rod **349** extends and retracts, because rod is connected to linkage **343**. More specifically, a distal end of rod **349** is rotatably connected to a side (such as a lateral outboard side) of linkage **343** about pivot axis **352**. As rod **349** extends and retracts, rod **349** causes linkage **343** to rotate with pivot bar **341** about pivot axis **345**. As will be explained (below), engagement rod can engage with and thereby contact a surface (a bottom surface) of each respective knife **129** and thereby move (that is, forcibly insert and forcibly retract relative to slots **232**) the knives **129** altogether as a group. Alternatively, engagement apparatus **340** can be formed in segments—each segment including each element of engagement apparatus **340**—such that groups of knives **129** less than the total number of knives **129** can be moved together as a group. Alternatively, engagement apparatus **340** can be formed in segments—each segment including each element of engagement apparatus **340**—such that each segment corresponds to a single knife **129** and individually moves knives **129**.

(23) Referring now to FIG. **4**, there is shown schematically a side view of cutting assembly **128**, with portions broken away. Cutting assembly **128**, in FIG. **4**, is shown to include floor **131** (with portions broken away, such that only a top surface of floor **131** is shown), knife **129** (only one knife **129** being visible in FIG. **4**), pivot **451**, engagement apparatus **340**, and knife overload protection mechanism **455**. Floor **131** includes a slot **232** through which knife **129** extends in FIG. **4** at least partially. Knife **129** includes a top surface which generally faces and thus encounters crop material **136** moving in flow direction **237**. Though not shown, this top surface can include a sharpened edge(s), serrations, and/or sharpened teeth configured for cutting through crop material **136**. Knife **129** is shown in its normal operating position, that being an insertion position. That is, knife **129** extends partially up through slot **232** so that the sharp top surface of knife can encounter and thus cut crop material **136**. Pivot **451** (which is coupled with frame **130**, as shown schematically in FIG. **4**) forms a pivot about which knife **129** can move and which knife **129** receives by way of a left-most mouth or cutout. Pivot **451** can be a bar, rod, or tube that extends substantially the transverse extent of floor **131** (like pivot bar **341**) and can have any cross-section and/or positioning that allows for knives **129** to be removed during servicing; for example, when in their insertion position **562**, knives **129** can be lifted out, or otherwise pivoted and lifted out, of the respective slot **232** and removed altogether from baler **101**. Engagement mechanism **340**, in FIG. **4**, is shown to include pivot bar **341** (which pivots about pivot axis **345**), linkage **343** which pivots with pivot bar **341**, engagement rod **342**, linear actuator **344** including cylinder **348** and rod **349**, cylinder **348**

being able to pivot about pivot axis 350, and rod 349 being able to pivot about pivot axis 352. Engagement apparatus 340 is in its home position. That is, rod 349 is fully retracted, and engagement rod 342 is not engaged and thus not in contact with knife 129. In this home position, linkage 343 can be said to point just above the 9 o'clock position (for example, at about 275 degrees, with 0 degrees being directly upward, that is, the 12 o'clock position). This home position is the normal operating position of engagement apparatus. For, engagement apparatus is primarily used to clear dirt and debris from slot 232 and/or knife 129 when knife 129 has become stuck and thus unable either to penetrate up through slot 232 so as to occupy the insertion position or to retract back down through slot 232 so as to occupy a retraction position. Thus, in normal operation of cutting assembly 128, engagement mechanism 340 is not needed.

(24) Regarding the knife overload protection mechanism 455, this mechanism 455 includes rod 464, a linkage 457, an engagement shaft 458, and a spring 459. Though fixing mechanism 460 provides functionality for mechanism 455 and is coupled with frame 130, fixing mechanism 460 is not included in mechanism 455, as it can serve a series of mechanisms 455 assigned to a series of individual knives 129. Rod 464 extends through a hole in fixing mechanism 460 and is coupled with linkage 457, such as by a bolt, pin, or welding. Fixing mechanism 460 can be formed as a bracket 460 and is coupled with frame 130, as schematically shown in FIG. 4 (bracket 460 can be smaller than what is shown in the figures). Linkage 457 can be connected at its other end to engagement shaft 458, which can engage a downstream surface of knife 129. Engagement shaft 458 is normally in a notch of knife 129, as shown in FIGS. 4 and 5. Spring 459 can surround and mount to rod 464 (as shown) and be braced at both ends to limit the extent of spring 459, being limited at its left end (as viewed in FIG. 4) by a ring fixed about the rod 464 and being limited at its right end by bracket 460 (another embodiment of the combination of rod 464 and spring 459 can provide that rod 464 has two parallel sides along its longitudinal extent and a reduced height portion along its longitudinal extent beginning near its distal end (opposite bracket 460) and running to its proximal end, with spring 459 surrounding this reduced height portion). Spring 459 can provide a compressive force to linkage 457, which biases knife 129 in the up position (that is, its insertion position 562). Mechanism 455 is specific to each knife 129, such that mechanism 455 is individually associated with a single knife 129; stated another way, a single one of overload protection mechanism 455 is assigned to and coupled with a single one of the knife 129. Thus, each knife 128 has its own overload protection mechanism 455. An actuator is not attached to mechanism 455 so as to push knife up or down, for example, so as to clear knife 129 of jams caused by dirt and debris relative to floor 131. Overload protection mechanism 455 protects each knife from an overload situation, such as when a rock is taken up by the pickup 111 and fed in direction of travel 237 to floor 131, with the result that this rock can strike one or more knives 129, which can destroy or otherwise damage and thus hinder the functionality of the knives 129 that are struck. By having mechanism 455 assigned to only one knife 129, only those knives that are struck by a foreign object will give way and thus retract under the force of the foreign object, rather than all of the knives, thereby allowing cutting to continue by those knives 129 that were not struck by the foreign object. In normal operation, the compressive force of spring 459, together with the positioning of engagement shaft 458 in the notch of the downstream side of knife 129, urges knife 129 in the insertion (up) position, as shown in FIG. 4. As indicated, when a knife 129 encounters the foreign object, knife 129 can retract momentarily all the way below the surface of floor 131 (alternatively, the retraction of knife 129 can be such that knife 129 does not completely retract under floor 131, but partially). When such retraction occurs, engagement shaft 458 can exit the notch and ride up a downstream side of knife 129 as spring compresses (like in FIG. 6, though FIG. 6 corresponds to when engagement mechanism 340 forces knife 129 to retract, not when a foreign object causes knife 129 to retract). Then, during normal operation, once the foreign object passes by knife 129, spring 459 causes engagement shaft 458 of the individual knife overload protection mechanism 455 to press on knife 129 and thereby to cause knife 129 to spring back up into place,

that is, into insertion position **562**, and engagement shaft **458** can slide back down knife **129** and return to the notch (as shown in FIG. **4**). This movement of knife **129** (retraction, and return to insertion position **562** under the force of spring **459**) due to a foreign object (and removal of the foreign object) occurs only when engagement mechanism **340** is not engaged with knife **129** (as in FIG. **4**); only then can the individual knife overload protection mechanism **455** perform its primary function of allowing the individual knife **129** to retract and then forcing the knife **129** back to insertion position **562**. This ability of knife **129** to retract under the influence of a foreign object and to re-insert under the influence of overload protection mechanism **455** is shown by double-arrow **461**, in FIG. **4**.

(25) Further, knife overload protection mechanism **455** is spaced apart from engagement mechanism **340**. As shown in FIG. **4**, knife overload protection mechanism **455** is positioned at a downstream end or portion of knife **129**, and engagement mechanism **340** is positioned generally at an upstream end or portion of knife **129**. Knife overload protection mechanism **455** and engagement mechanism **340** are discrete and distinct from one another. That is, overload protection to each knife **129** is provided by way of overload protection mechanism **455** apart from engagement mechanism **340**, and the upward force for insertion and the downward force for retraction of knives **129** is provided by engagement mechanism **340**. Thus, by way of overload protection mechanism **455**, overload protection is provided to each knife **129** individually (independent of the other knives **129**); and, by way of engagement mechanism **340**, knives **129** can be forcibly moved and locked all at the same time (inserted or retracted) in order to service the knives, clear dirt and debris, or to remove knives **129** from operation without removing them from cutting assembly **128** (that is, moving knives **129** and locking them in a retracted position below floor **131**).

(26) Referring now to FIG. **5**, there is shown a view similar to FIG. **4**, with substantially similar structure. The primary difference between FIGS. **4** and **5** is that engagement apparatus **340** has moved from its home position (FIG. **4**) to a first position (FIG. **5**). Correspondingly, engagement mechanism **340** is configured for selectively engaging with knife **129** and thereby for forcing knife **129** (positively pushing knife **129**) to occupy a first position, which corresponds to the first position of engagement mechanism **340**. As shown in FIG. **5**, this first position **562** of knife **129** can be the insertion position **562**, that is, a fully inserted position of knife. Further, when knife **129** is in this insertion position **562** by way of engagement apparatus **340**, knife **129** is locked into this insertion position **562** by way of engagement apparatus **340**. That is, knife **129** is not free to fall back down through slot **232**, because engagement apparatus **340** is pushing knife **129** up and engagement apparatus **340** is held in position, until adjusted. In operation, when engagement apparatus **340** is in its home position (FIG. **4**), cylinder **348** can receive hydraulic fluid, causing rod **349** to extend, which causes pivot bar **341** to rotate clockwise (as viewed in FIG. **5**), until engagement rod **342** is approximately in the 12 o'clock position. In this movement, from the home position to the first position of the engagement apparatus **340**, engagement rod **342** engages and thus comes into contact with a bottom surface **453** of knife **129** and can forcibly raise knife **129** into its insertion position **562**, as shown in FIG. **5**. For example, knife **129** could be stuck below or partially below floor **131** by dirt or debris **454**, and the forceful push of knife **129** by engagement rod will push knife **129** through this dirt/debris **454** (thereby clearing a jam caused by dirt/debris **454**) and thus through slot **232** so as to be fully inserted. When engagement apparatus **340** is in its first position, this is not a normal operating configuration for cutting assembly **128**, and baler **101** should not be operated across a field in this manner; for, this position locks knives **129** into their insertion position **562** and thus unable to give way—under the influence of overload protection mechanism **455**—when struck by a foreign object. Using engagement apparatus **340** to force knife **129** into insertion position **562** enables the clearing of dirt and debris, and allows the user to remove the knives from their slots for servicing or replacement. From this first position of engagement apparatus **340**, engagement mechanism **340** has two options (beside remaining in this position).

First, engagement mechanism **340** is configured for selectively disengaging from knife **129** and returning to its home position (FIG. **4**). In so doing, rod **349** retracts and linkage **343** pivots counter-clockwise (as viewed in FIG. **5**). Second, engagement apparatus **340** is configured for moving clockwise and thus for selectively engaging knife **129** and thereby for forcing knife **129** to occupy a second position **663**, namely, a retraction position **663**, which is also locked by engagement apparatus **340**. In proceeding to this second position **663**, thus forcing knives **129** fully up in their first position **562** and fully down in their second position **663**. This will provide a full “clean out,” wherein a clean-out cycle can be deemed to begin at the home position of engagement apparatus **340**, proceed to the first position of engagement apparatus **340** (corresponding to first position **562**), then to the second position of engagement apparatus **340** (corresponding to second position **663**), then back to the first position of engagement apparatus again (clearing out any remaining dirt or debris), and then back to the home position of engagement apparatus **340**. This second option **663** is discussed in reference to FIG. **6**. Alternatively, engagement apparatus **340** can be configured to move from its home position (FIG. **4**) all the way to its second position (FIG. **6**) without stopping or otherwise pausing at its first position (FIG. **5**).

(27) Referring now to FIG. **6**, there is shown a view similar to FIGS. **4** and **5**, in that substantially similar structure is employed. The primary difference between FIGS. **5** and **6** is that engagement apparatus **340** has moved from its first position (its insertion position)(FIG. **5**) to its second position (its retraction position). Correspondingly, engagement mechanism **340** is configured for selectively engaging with knife **129** and thereby for forcing knife **129** (positively pulling knife **129**) to occupy the second position **663**, which corresponds to the second position of engagement mechanism **340**. As shown in FIG. **6**, this second position **663** of knife **129** can be the retraction position **663**, that is, a fully retracted position of knife **129** (though retraction position **663** is shown (in FIGS. **6-7**) such that knife **129** is fully below floor **131**, this need not be the case; rather, the retraction position can be with knife **129** partially below floor **131**). Further, when knife **129** is in this retraction position **663** by way of engagement apparatus **340**, knife **129** is locked into this retraction position **663** by way of engagement apparatus **340**. That is, knife **129** is not free to move back up through slot **232** (for instance, under the influence of knife overload protection mechanism **455**), because engagement apparatus **340** is forcibly pulling knife **129** down and engagement apparatus **340** is held in position, until adjusted. In operation, when engagement apparatus **340** is in its first position (FIG. **5**), cylinder **348** can receive hydraulic fluid, causing rod **349** to extend even further (for instance, to its fullest extent), which causes pivot bar **341** to rotate clockwise (as viewed in FIG. **6**), until engagement rod **342** is approximately in the 2 o'clock position. In this movement, from the first position to the second position of the engagement apparatus **340**, engagement rod **342** engages and thus comes into contact (or, remains in contact) with bottom surface **453** of knife **129** (for example, at the curve or notch of bottom surface **453** which is adjacent to linkage **343** in FIG. **6**) and can forcibly lower knife **129** into its retraction position **663**, as shown in FIG. **6**. For example, knife **129** could be stuck fully or partially above floor **131** by dirt or debris **454**, and the forceful pull of knife **129** by engagement rod will pull knife **129** through this dirt/debris **454** and thus through slot **232** so as to be fully retracted. When engagement apparatus **340** is in its second position, this need not be but can be an operating configuration of cutting assembly **128**. That is, baler **101** can be run across a field and bale with knives **129** in their forcibly retracted position **663** (which may be advantageous for certain harvesting conditions), though this is not necessarily the normal operating configuration for cutting assembly **128**. Using engagement apparatus **340** to force knife **129** into retraction position **663** enables the clearing of dirt and debris that may be jamming knife **129** in the insertion position **562**, and/or allows the user to run the baler with knives retracted into retraction position **663**. From this second position of engagement apparatus **340**, engagement mechanism **340** (besides remaining in this position) can selectively return back to its first position (FIG. **5**) or its home position (FIG. **4**) by retracting rod **349** and thus causing pivot bar **341** and thus linkage **343** to pivot counter-clockwise (as viewed in FIG. **6**).

(28) FIG. 6 further includes controller **123** and sensors **638**, **639** operatively coupled with controller **123** (and also controller **115**)(for illustrative purposes controller **123** and sensors **638**, **639** are shown in FIG. 6 but not FIGS. 4 and 5 as well, though they can be impliedly present). As indicate above, engagement apparatus **340** can be moved to different locations. For example, engagement apparatus **340** can be moved from its home position, to its first position, to its second position, back to its first position, and then back to its home position. Further, any other combination of movements can occur as well, for example, from the home position directly to the second position, without pausing at the first position, or the like. Engagement apparatus **340** can be so moved manually or by way of a control system, such by way of control system **122** of baler **101**. As described above, control system **122** includes controller **123**, and can include one or more sensors, such as sensor **638** and/or **639**, which are shown schematically (sensors can also be considered to be a part of control system **114**). Sensors **638**, **639** can be position sensors. Sensor **638** can be coupled with cylinder **348** (on an interior or an exterior of cylinder **348**) and can be configured for sensing a position of rod **349** relative to cylinder **348**, thus sensing the position of rod **349** in terms of its stroke. Sensor **639** can be coupled with linkage **343** and can be configured for sensing a position of linkage **343**, a position of pivot bar **341**, and/or a position of engagement rod **342**. Position data sensed by sensors **638**, **639** can be formed into position signals by sensors **638**, **639** and outputted by sensors **638**, **639** to controller **123**, which is configured for receiving the position signals from sensors **638**, **639**. Upon receiving these position signals, controller **123** is configured for outputting this information to a display of input/output device **120** in cab **105** of tractor **100**, so that user can know this information. In this way, user can be informed about the position of engagement apparatus **340**, whether it is in its home position, its first position, or its second position and thus can know whether knives **129** are in their forced upward position **562** (first/insertion position) or their forced downward position **663** (second/retraction position). Further, these sensors **638**, **639** or another sensor(s) can be configured for sensing a position of knives **129**, regardless of the position of engagement apparatus **340**, so as to inform the user whether knives **129** (individual, or as groups, or as the entire set of knives **129**) are fully inserted, fully retracted, or at some intermediate position therebetween.

(29) Further, a user (such as the operator of tractor **100**), upon being informed of the position of engagement apparatus **340** and/or knives **129** by way of input/output device **120**, can issue a command to controller **123** by way of device **120** to move engagement apparatus **340** and thus knives **129** to a certain position, for example, to any of the positions of engagement apparatus **340** described above, namely, its home position, its first position, and its second position (or any intermediate position thereof). For example, user can issue a command to move engagement apparatus **340** from its home position to its first position (corresponding to insertion position **562** of knives **129**). Upon doing so, controller **123** is configured for outputting a controller output signal to actuator **344**, for example, to extend rod **349** so that linkage **343** pivots to approximately the 12 o'clock position. As sensors **638**, **639** sense the position of rod **349**, linkage **343**, pivot bar **341**, and/or engagement rod **342**, controller **123** can be configured so as to know when to output a stop signal to actuator **344** so as to halt movement of engagement rod **342** at the first position of engagement apparatus **340**, such that knives **129** are in their insertion position **562**. Likewise, upon user command, controller **123** can output a signal so that actuator **344** further extends rod **349** to move engagement rod **342** to the second position of engagement apparatus **340**, and to output a stop signal to halt actuator **344** when sensors **638**, **639** sense the position of rod **349**, linkage **343**, pivot bar **341**, and/or engagement rod **342**, corresponding to when engagement rod **342** reaches the second position of engagement apparatus **340** and knives are in their corresponding retraction position **663**. Further, a command can be issued to return engagement apparatus to its home position, and further commands can be issued by user, such that controller **123** and sensors **348**, **349** act in corresponding ways, depending upon the specific command.

(30) Referring now to FIG. 7, there is shown an alternative embodiment of the cutting assembly of

the present invention, now labeled cutting assembly **728**. FIG. 7 is similar to FIGS. 4-6, both structurally and functionally. Substantially similar structure, structurally and functionally, has the same reference character, or a reference character that is raised by a factor of 100, and will not be described again here, unless otherwise noted. The primary difference with FIG. 7 relative to prior figures is the engagement apparatus, now labeled engagement apparatus **740**. Engagement apparatus **740** includes pivot bar **341**, joining mechanism **743**, engagement rod **342**, and actuator **744**. Pivot bar **341** includes pivot axis **345** and can pivot in either direction about pivot axis **345**. Joining mechanism **743** can be generally formed as a disc **743**, which functions similarly to linkage **343**. Disc **743** can be made of any suitable material, such as steel. Two discs **743** can be attached respectively to lateral ends of pivot bar **341** and lateral ends of engagement rod **342**, the attachments being by welding or pinning, or the like. Disc **743** pivots with pivot bar **341** and thus about pivot axis **345**. For illustrative purposes, in dash-dot-dot lines, a full 360 degree rotation of disc **743** (with respect to an outer perimeter of disc **743**) about pivot axis **345** is shown; though a 360 degree path is shown, it can be appreciated that disc **743** may be limited to pivoting about a lesser angular extent, for example, less than 180 degrees, such as 170 degrees (such as just above the 9 o'clock position to approximately the 2 o'clock position, by way of the 12 o'clock position). Engagement rod **342** engages a bottom surface **453** of knife **129**, as described above. Actuator **744** is shown schematically and, in this embodiment, is formed as a rotary actuator **744**. Rotary actuator **744** can be, for example, an electric motor or a hydraulic motor with a rotatable output shaft which can be connected to pivot bar **341** in any suitable manner, such as by a splined connection coaxial with pivot bar **341** (to either side of disc **743**, for example), or by a belt(s), a chain(s), and/or gearing. Control system **122** of baler **101** (and control system **114**) can further include a sensor **739**, which is a position sensor and which is operatively coupled with, and thus communicates with, controller **123** (and controller **115**); sensor **739** functions substantially similarly to sensors **638**, **639**, unless otherwise noted. Sensor **739** can be attached pivot bar **741**, disc **743**, or other suitable structure, so as to be able to detect a position of pivot bar **741** and provide this position data in the form of position signals to controller **123**. FIG. 7 supplies similar information to what is provided in FIGS. 4-6 with respect to the first embodiment of the cutting assembly. That is, FIG. 7 shows, in solid lines, engagement rod **342** in substantially the 6 o'clock position (similar to FIG. 5) and thus the first position of engagement apparatus **740**, with knife **129** in the insertion position **562**. Near the 9 o'clock position (similar to FIG. 4), engagement rod **342** is shown in dashed lines, to indicate the home position of engagement apparatus **740**. Near the 2 o'clock position (similar to FIG. 6), engagement rod **342** is shown again in dashed lines, to indicate the second position of engagement apparatus **740**, which corresponds to the retraction position **663** of knife **129**, shown in dash-dot-dot lines.

(31) In use, a user of tractor **100** and baler **101** can bale crop material **136** (such as baling hay), using a cutting assembly **128**, **728** as part of feeder system **108**. Cutting assembly **128**, **728** has individual knife overload protection, by way of knife overload protection mechanism **455**. Further, cutting assembly **128**, **728** includes a way to clear jams of knives **129** due to dirt and debris **454**, to otherwise service one or more knives **129**, and/or to run baler **101** with knives **129** forcible retracted and locked in retraction position **663**. In normal operation of baler **101**, user can run with knives **129** up and in their insertion position, held up by knife overload protection mechanism **455**, but not in any way by engagement apparatus **340**, **740**. If a sensor detects, for example, that one or more knives **129** are jammed by dirt and debris from being able to insert fully up through slots **232**, or user otherwise wishes to ensure clearance of any possible dirt and debris, user can issue a command by way of input/output device **120** to cause control system **122** to move engagement apparatus **340**, **740** from its home position to its first position (6 o'clock position), so as to forcibly push knives **129** through any dirt and debris block slots **232** and to lock knives in the insertion position **562**. If user wishes to complete the clean-out or to otherwise retract knives **129** down to retraction position **663**, user can issue a command to move engagement apparatus **340**, **740** to its

second position (2 o'clock position). User can also command engagement apparatus **340, 740** to return to its home position, the engagement apparatus **340** no longer being engaged with knives **129**, though knife overload protection mechanism **455** maintains its engagement with individual knives **129**. Thus, in accordance with the present invention, the user has positive engagement with knives **129** to raise or lower them on command, while maintaining individual overload protection by way of overload protection mechanism **455** using springs **459** as part of the individual overload protection mechanism **455**.

(32) Referring now to FIG. **8**, there is shown a flow diagram showing a method **865** for using an agricultural baler **101**, the method including the steps of: providing **866** a frame **130** and a feeder system **108** coupled with the frame **130**, the feeder system **108** including a cutting assembly **128, 728** coupled with the frame **130** and including at least one knife **129** and an engagement apparatus **340, 740**, the at least one knife **129** configured for cutting a crop material **136**, the engagement apparatus **340, 740** being configured for being spaced apart from at least one overload protection mechanism **455** individually associated with a single one of the at least one knife **129**; engaging **867** selectively, by way of the engagement apparatus **340, 740**, with the at least one knife **129** and thereby forcing the at least one knife **129** to occupy a first position **562**; and engaging **868** selectively, by way of the engagement apparatus **340, 740**, with the at least one knife **129** and thereby forcing the at least one knife **129** to occupy a second position **663**. The cutting assembly **128, 728** can further include the at least one overload protection mechanism **455**, a single one of the at least one overload protection mechanism **455** being assigned to and coupled with a single one of the at least one knife **129**. The method **865** can further include the step of disengaging **869** selectively, by way of the engagement apparatus **340, 740**, from the at least one knife **129**. The first position **562** can be an insertion position **562**, and the second position **663** can be a retraction position **663**, both the first position **562** and the second position **663** being locked. The at least one knife **129** can include a surface **453**, the engagement apparatus **340, 740** including an actuator **344, 744** and an engagement rod **342** configured for engaging the surface **453** of the at least one knife **129**. The actuator **344, 744** can be linear actuator **344** or a rotary actuator **744**.

(33) It is to be understood that the steps of method **865** are performed by controller **115, 123** upon loading and executing software code or instructions which are tangibly stored on a tangible computer readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by controller **115, 123** described herein, such as the method **865**, is implemented in software code or instructions which are tangibly stored on a tangible computer readable medium. The controller **115, 123** loads the software code or instructions via a direct interface with the computer readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by controller **115, 123**, controller **115, 123** may perform any of the functionality of controller **115, 123** described herein, including any steps of the method **865**.

(34) The term “software code” or “code” used herein refers to any instructions or set of instructions that influence the operation of a computer or controller. They may exist in a computer-executable form, such as machine code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term “software code” or “code” also includes any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.

(35) These and other advantages of the present invention will be apparent to those skilled in the art from the foregoing specification. Accordingly, it is to be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing

from the broad inventive concepts of the invention. It is to be understood that this invention is not limited to the particular embodiments described herein, but is intended to include all changes and modifications that are within the scope and spirit of the invention.

Claims

1. A feeder system of an agricultural baler, the feeder system being coupled with a frame of the agricultural baler, the feeder system comprising: a cutting assembly coupled with the frame and including: at least one knife configured for cutting a crop material; an engagement apparatus configured for: being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife, the at least one overload protection mechanism engaging with a rearward surface of the single one of the at least one knife; selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a first position, the engagement apparatus engaging with another a bottom surface of the single one of the at least one knife; and selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a second position.
2. The feeder system of claim 1, wherein the cutting assembly further includes the at least one overload protection mechanism, the engagement apparatus being spaced apart from the at least one overload protection mechanism, a single one of the at least one overload protection mechanism being assigned to and coupled with a single one of the at least one knife.
3. The feeder system of claim 2, wherein the engagement apparatus is further configured for selectively disengaging from the at least one knife.
4. The feeder system of claim 3, wherein the first position is an insertion position, and the second position is a retraction position, both the first position and the second position being locked.
5. The feeder system of claim 4, wherein the at least one knife includes a surface, the engagement apparatus including an actuator and an engagement rod configured for engaging the surface of the at least one knife.
6. The feeder system of claim 5, wherein the actuator is one of a linear actuator and a rotary actuator.
7. An agricultural baler, comprising: a frame; a feeder system coupled with the frame and including: a cutting assembly coupled with the frame and including: at least one knife configured for cutting a crop material; an engagement apparatus configured for: being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife, the at least one overload protection mechanism engaging with a rearward surface of the single one of the at least one knife; selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a first position, the engagement apparatus engaging with a bottom surface of the single one of the at least one knife; and selectively engaging with the at least one knife and thereby for forcing the at least one knife to occupy a second position.
8. The agricultural baler of claim 7, wherein the cutting assembly further includes the at least one overload protection mechanism, the engagement apparatus being spaced apart from the at least one overload protection mechanism, a single one of the at least one overload protection mechanism being assigned to and coupled with a single one of the at least one knife.
9. The agricultural baler of claim 8, wherein the engagement apparatus is further configured for selectively disengaging from the at least one knife.
10. The agricultural baler of claim 9, wherein the first position is an insertion position, and the second position is a retraction position, both the first position and the second position being locked.
11. The agricultural baler of claim 10, wherein the at least one knife includes a surface, the engagement apparatus including an actuator and an engagement rod configured for engaging the surface of the at least one knife.
12. The agricultural baler of claim 11, wherein the actuator is one of a linear actuator and a rotary

actuator.

13. A method for using an agricultural baler, the method comprising the steps of: providing a frame and a feeder system coupled with the frame, the feeder system including a cutting assembly coupled with the frame and including at least one knife and an engagement apparatus, the at least one knife configured for cutting a crop material, the engagement apparatus being configured for being spaced apart from at least one overload protection mechanism individually associated with a single one of the at least one knife, the at least one overload protection mechanism engaging with a rearward surface of the single one of the at least one knife; engaging selectively, by way of the engagement apparatus, with the at least one knife and thereby forcing the at least one knife to occupy a first position, the engagement apparatus engaging with a bottom surface of the single one of the at least one knife; and engaging selectively, by way of the engagement apparatus, with the at least one knife and thereby forcing the at least one knife to occupy a second position.

14. The method of claim 13, wherein the cutting assembly further includes the at least one overload protection mechanism, the engagement apparatus being spaced apart from the at least one overload protection mechanism, a single one of the at least one overload protection mechanism being assigned to and coupled with a single one of the at least one knife.

15. The method of claim 14, further including the step of disengaging selectively, by way of the engagement apparatus, from the at least one knife.

16. The method of claim 15, wherein the first position is an insertion position, and the second position is a retraction position, both the first position and the second position being locked.

17. The method of claim 16, wherein the at least one knife includes a surface, the engagement apparatus including an actuator and an engagement rod configured for engaging the surface of the at least one knife.

18. The method of claim 17, wherein the actuator is one of a linear actuator and a rotary actuator.
